BIO 102 GENERAL BIOLOGY 2

BASIC CHARACTERISTICS, IDENTIFICATION AND CLASSIFICATION OF BACTERIA WITH EXAMPLES

Introduction

Bacteria are single-celled microorganisms that exist in diverse shapes, sizes, and environments. We have studied their basic characteristics in the previous module (BIO 101). These include their cell wall structure, reproduction, metabolism, genetic diversity, and ecology. In this module, we shall look into how to identify and classify bacteria with examples.

IDENTIFICATION OF BACTERIA

Sample Collection and Preparation

- Collect samples from various sources (e.g., soil, water, clinical specimens) using sterile techniques.
- Sample is prepared by diluting, if necessary, and plating it on appropriate growth medium

Common sample types include:

1. **Clinical Samples**: These are collected from patients and can include:
    - Blood
    - Urine
    - Sputum
    - Cerebrospinal fluid
    - Tissue biopsies
    - Swabs from infected sites (e.g., throat, wound, nasal)
2. **Environmental Samples**: These are collected from natural or artificial environments and can include:
    - Soil
    - Water (rivers, lakes, oceans, drinking water)
    - Air
    - Surfaces in healthcare settings (e.g., hospital rooms, equipment)
3. **Food Samples**: These are used in the food industry to ensure safety and quality, and can include:
    - Raw meats
    - Dairy products
    - Fruits and vegetables
    - Processed foods
4. **Industrial Samples**: These are collected from industrial environments to monitor contamination and can include:
    - Biofilms in pipes and reactors
    - Fermentation products
    - Industrial waste

5. **Research Samples**: These can come from various sources depending on the focus of the research and can include:
    - Model organisms (e.g., mice, plants)
    - Synthetic constructs (e.g., genetically modified bacteria)
    - Environmental DNA (eDNA) samples for metagenomic studies

Methods of examination

1. **Culture-Based Methods**

- **Selective and Differential Media**: Culturing bacteria on specialized media allows for the selective growth of certain bacterial species based on specific nutritional requirements or metabolic characteristics.

2. **Macroscopic Examination**

- **Colony Morphology**: Observing the appearance of bacterial colonies on agar plates can provide initial clues about the identity of bacteria. Characteristics such as size, shape, color, texture, and margin are noted.

3. **Microscopic Examination**

- **Gram Staining**: One of the most widely used staining techniques to differentiate bacteria based on their cell wall structure. Gram-positive bacteria retain the crystal violet stain, appearing purple under the microscope, while gram-negative bacteria appear pink after counterstaining with safranin.
- **Cell Morphology**: Microscopic examination allows the observation of bacterial cell shape, arrangement, and presence of specific structures such as flagella or capsules.

4. **Biochemical Tests**

Various biochemical tests are performed to assess the metabolic capabilities of bacteria. Common tests include:

- **Catalase Test**: Differentiates bacteria based on their ability to produce the enzyme catalase. Positive: *Staphylococcus*, Negative: *Streptococcus*
- **Oxidase Test**: Detects the presence of cytochrome c oxidase in bacteria. Positive: *Pseudomonas aeruginosa*, Negative: *Escherichia coli*
- **Indole Production Test**: Determines the ability of bacteria to produce indole from tryptophan. Positive: *Escherichia coli*, Negative: *Enterobacter cloacae*
- **Citrate Utilization Test**: Determines the ability to use citrate as the sole carbon source. Positive: *Klebsiella pneumoniae*, Negative: *Escherichia coli*
- **Coagulase Test:** To differentiate *Staphylococcus aureus* (coagulase-positive) from other *Staphylococcus* species (coagulase-negative).
- **Methyl Red (MR) Test**: Detects the production of stable acids from glucose fermentation. Positive: *Escherichia coli*, Negative: *Enterobacter aerogenes*
- **Sugar Fermentation Tests**: Assess the ability of bacteria to ferment specific sugars, producing characteristic metabolic byproducts.

5. **Molecular Techniques**

- **Polymerase Chain Reaction (PCR)**: Amplifies specific regions of bacterial DNA for identification purposes. Techniques such as:
    - **16S rRNA Sequencing**: Targets the highly conserved 16S ribosomal RNA gene for phylogenetic analysis and bacterial identification.
    - **Whole Genome Sequencing (WGS)**: Provides comprehensive genetic information for accurate identification and characterization of bacteria.

6. **Serological Methods**

- **Immunological Assays**: Serological techniques involve the use of antibodies to detect specific antigens or antibodies produced by bacteria. Examples include:
    - **Enzyme-Linked Immunosorbent Assay (ELISA)**: Detects bacterial antigens or antibodies in patient samples.
    - **Western Blotting**: Identifies specific proteins within bacterial extracts using antibodies.

7. **Anaerobic Culture Techniques**: Some bacteria require anaerobic conditions for growth, necessitating specialized culture techniques such as anaerobic chambers or specific

Types of Bacterial Isolation Media

1. **General Purpose Media**

    Nutrient Agar (NA)

    - Composition: Peptone, beef extract, sodium chloride, agar.
    - Use: Supports the growth of a wide range of non-fastidious bacteria.
    - Example: Used for routine culturing and sub-culturing of bacteria.

2. **Enriched Media**
    - **Blood Agar (BA)**
        - Composition: Nutrient agar base supplemented with 5-10% sheep or horse blood.
        - Use: Supports the growth of fastidious bacteria and differentiates bacteria based on hemolytic properties.
        - Example: *Streptococcus pyogenes* (beta-hemolysis), *Streptococcus pneumoniae* (alpha-hemolysis).
    - **Chocolate Agar (CHOC)**
        - Composition: Nutrient agar with lysed red blood cells.
        - Use: Supports the growth of fastidious respiratory bacteria.
        - Example: *Haemophilus influenzae*, *Neisseria gonorrhoeae*.
3. **Selective Media**
    - **MacConkey Agar (MAC)**
        - Composition: Peptone, lactose, bile salts, crystal violet, neutral red, agar.

        - Use: Selective for Gram-negative bacteria and differentiates lactose fermenters from non-fermenters.
        - Example: *Escherichia coli* (lactose fermenter, pink colonies), *Salmonella* spp. (non-lactose fermenter, colorless colonies).
    - **Mannitol Salt Agar (MSA)**
        - Composition: Peptone, beef extract, mannitol, sodium chloride, phenol red, agar.
        - Use: Selective for staphylococci due to high salt concentration; differentiates mannitol fermenters.
        - Example: *Staphylococcus aureus* (mannitol fermenter, yellow colonies), *Staphylococcus epidermidis* (non-mannitol fermenter, red colonies).
4. **Differential Media**
    - **Eosin Methylene Blue Agar (EMB)**
        - Composition: Peptone, lactose, eosin Y, methylene blue, agar.
        - Use: Differentiates lactose fermenters, which produce dark colonies with metallic sheen.
        - Example: *Escherichia coli* (dark colonies with green metallic sheen), *Enterobacter aerogenes* (pink colonies).
    - **Xylose Lysine Deoxycholate Agar (XLD)**
        - Composition: Yeast extract, xylose, lysine, lactose, sucrose, deoxycholate, phenol red, agar.
        - Use: Selective and differential for enteric pathogens, especially *Salmonella* and *Shigella*.
        - Example: *Salmonella* spp. (red colonies with black centers), *Shigella* spp. (red colonies).
5. **Specialized Media**
    - **Thiosulfate-Citrate-Bile Salts-Sucrose Agar (TCBS)**
        - Composition: Peptone, yeast extract, bile salts, citrate, thiosulfate, sucrose, bromothymol blue, thymol blue, agar.
        - Use: Selective for *Vibrio* species.
        - Example: *Vibrio cholerae* (yellow colonies due to sucrose fermentation), *Vibrio parahaemolyticus* (green colonies).
    - **Lowenstein-Jensen Medium (LJ)**
        - Composition: Eggs, glycerol, malachite green, potato flour, mineral salts.
        - Use: Selective for *Mycobacterium* species.
        - Example: *Mycobacterium tuberculosis* (growth of rough, buff-colored colonies).

CLASSIFICATION OF BACTERIA

Classifification based on Cell Wall Composition

- **Gram-positive Bacteria**
    - Example: *Staphylococcus aureus* - A spherical, Gram-positive bacterium known for causing skin infections, pneumonia, and food poisoning.
- **Gram-negative Bacteria**

- Example: *Escherichia coli* - A rod-shaped, Gram-negative bacterium commonly found in the intestines of humans and other warm-blooded animals. While most strains are harmless, some can cause foodborne illness and urinary tract infections.

Classification based on Morphology

- **Cocci (Spherical Bacteria)**
  - **Monococcus**: Single spherical cell.
    - Example: *Micrococcus luteus*
  - **Diplococci**: Pairs of spherical cells.
    - Example: *Neisseria gonorrhoeae* (causes gonorrhea)
  - **Streptococci**: Chains of spherical cells.
    - Example: *Streptococcus pyogenes* (causes strep throat)
  - **Staphylococci**: Clusters of spherical cells, resembling grapes.
    - Example: *Staphylococcus aureus* (can cause skin infections)
  - **Tetrads**: Groups of four spherical cells.
    - Example: *Micrococcus tetragenus*
  - **Sarcinae**: Cubic configuration of eight spherical cells.
    - Example: *Sarcina ventriculi*
- **Bacilli (Rod-shaped Bacteria)**
  - **Single Bacillus**: Single rod-shaped cell.
    - Example: *Escherichia coli* (commonly found in the intestines)
  - **Diplobacilli**: Pairs of rod-shaped cells.
    - Example: *Coxiella burnetii* (causes Q fever)
  - **Streptobacilli**: Chains of rod-shaped cells.
    - Example: *Streptobacillus moniliformis* (causes rat-bite fever)
  - **Coccobacilli**: Short and plump rod-shaped cells.
    - Example: *Haemophilus influenzae* (can cause respiratory infections)
- **Spirilla (Spiral-shaped Bacteria)**
  - **Spirillum**: Rigid spiral-shaped cells.
    - Example: *Spirillum minus* (causes rat-bite fever)
  - **Spirochetes**: Flexible, tightly coiled spiral-shaped cells.
    - Example: *Treponema pallidum* (causes syphilis)
    - Example: *Borrelia burgdorferi* (causes Lyme disease)
- **Vibrios (Comma-shaped Bacteria)**
  - **Vibrio**: Curved rod-shaped cells, resembling a comma.
    - Example: *Vibrio cholerae* (causes cholera)
- **Pleomorphic Bacteria (Variable Shape)**
  - Bacteria that can alter their shape or size in response to environmental conditions.
    - Example: *Mycoplasma pneumoniae* (lacks a cell wall and causes atypical pneumonia)

4. Based on Oxygen Requirement

Bacteria can also be classified based on their oxygen requirements. Here are the main types, along with examples for each:

1. **Obligate Aerobes**
    - Require oxygen for growth as they use it for aerobic respiration.
    - Example: *Mycobacterium tuberculosis* (causes tuberculosis)
    - Example: *Bacillus subtilis* (common soil bacterium)
2. **Obligate Anaerobes**
    - Cannot survive in the presence of oxygen; they obtain energy through anaerobic respiration or fermentation.
    - Example: *Clostridium botulinum* (causes botulism)
    - Example: *Bacteroides fragilis* (part of the normal gut flora, but can cause infections)
3. **Facultative Anaerobes**
    - Can grow in the presence or absence of oxygen; they use aerobic respiration when oxygen is available and switch to anaerobic respiration or fermentation when it is not.
    - Example: *Escherichia coli* (commonly found in the intestines)
    - Example: *Staphylococcus aureus* (can cause skin infections)
4. **Microaerophiles**
    - Require oxygen at lower concentrations than atmospheric levels (typically 2-10%); higher concentrations of oxygen can be harmful to them.
    - Example: *Helicobacter pylori* (causes stomach ulcers)
    - Example: *Campylobacter jejuni* (causes food poisoning)
5. **Aerotolerant Anaerobes**
    - Do not require oxygen for growth but can tolerate its presence; they obtain energy through fermentation.
    - Example: *Lactobacillus* species (important in food fermentation processes, such as yogurt and sauerkraut)
    - Example: *Streptococcus pyogenes* (causes strep throat)

5. Based on Metabolic Characteristics

- **Autotrophic Bacteria**
    - Example: *Nitrosomonas europaea* - An autotrophic bacterium that oxidizes ammonia to nitrite as part of the nitrogen cycle.
- **Heterotrophic Bacteria**
    - Example: *Pseudomonas aeruginosa* - A versatile heterotrophic bacterium found in soil, water, and on plants, known for its opportunistic pathogenicity in humans, particularly in immunocompromised individuals.

6. Based on Ecological Roles

- **Symbiotic Bacteria**
  - Example: *Rhizobium leguminosarum* - A symbiotic bacterium that forms nitrogen-fixing nodules on the roots of leguminous plants, aiding in nitrogen fixation and enhancing plant growth.
- **Pathogenic Bacteria**
  - Example: *Salmonella enterica* - A pathogenic bacterium responsible for causing salmonellosis, a common foodborne illness characterized by symptoms such as diarrhea, fever, and abdominal cramps.

Systematic Classification of Bacteria

Systematic classification, also known as taxonomy, is the science of organizing living organisms into hierarchical categories based on their evolutionary relationships, genetic similarities, and shared characteristics. Bacteria, being one of the most diverse groups of organisms, are classified into several levels of hierarchy. Here's an overview of the systematic classification of bacteria with examples:

Domain: Bacteria

- Bacteria belong to the domain Bacteria, one of the three domains of life, along with Archaea and Eukarya.

Kingdom: Eubacteria

Phylum: Firmicutes

- **Class: Bacilli**
  - *Example*: *Bacillus subtilis* - A Gram-positive, rod-shaped bacterium commonly found in soil and the gastrointestinal tract of humans and animals. It's widely studied for its ability to produce antibiotics and other bioactive compounds.
- **Class: Clostridia**
  - *Example*: *Clostridium botulinum* - A spore-forming, anaerobic bacterium known for producing botulinum toxin, the most potent toxin known, causing botulism, a severe and potentially fatal paralytic illness.

Phylum: Proteobacteria

- **Class: Alphaproteobacteria**
  - *Example*: *Rhizobium leguminosarum* - A nitrogen-fixing bacterium that forms symbiotic relationships with leguminous plants, helping them convert atmospheric nitrogen into a form usable by the plants.
- **Class: Betaproteobacteria**
  - *Example*: *Neisseria gonorrhoeae* - A gram-negative diplococcus bacterium responsible for the sexually transmitted infection gonorrhea.
- **Class: Gammaproteobacteria**

    - *Example*: *Escherichia coli* - A facultatively anaerobic, gram-negative bacterium commonly found in the lower intestine of warm-blooded organisms. While most strains are harmless, some can cause severe foodborne illness.
- **Class: Deltaproteobacteria**
    - *Example*: *Desulfovibrio vulgaris* - A sulfate-reducing bacterium found in various environments, including soil, sediment, and the human gastrointestinal tract.

Phylum: Actinobacteria

- **Class: Actinobacteria**
    - *Example*: *Mycobacterium tuberculosis* - A rod-shaped, acid-fast bacterium responsible for causing tuberculosis in humans and other animals. It's one of the most significant bacterial pathogens worldwide.

Phylum: Cyanobacteria

- **Class: Cyanobacteria**
    - *Example*: *Anabaena sp.* - A filamentous, photosynthetic bacterium capable of nitrogen fixation. It forms symbiotic relationships with certain plants and is a prominent member of aquatic ecosystems.

Phylum: Spirochaetes

- **Class: Spirochaetes**
    - *Example*: *Borrelia burgdorferi* - A spiral-shaped, gram-negative bacterium responsible for causing Lyme disease, a multisystemic illness transmitted through the bite of infected ticks